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Experimental study of "$CuO_{0.5}$"-"FeO"-SiO_2 and "FeO"-SiO_2 systems in equilibrium with metal at 1400–1680 °C

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ABSTRACT

The phase equilibria of the "$CuO_{0.5}$"-"FeO"-SiO_2 slag system in equilibrium with metallic copper and tridymite/cristobalite (SiO_2), and the "FeO"-SiO_2 slag system in equilibrium with metallic iron and tridymite/cristobalite (SiO_2) were investigated between 1400 and 1680 °C. High temperature equilibration in closed silica ampoules, followed by quenching and the direct measurement of the phase compositions with the electron probe X-ray microanalysis (EPMA) was used. The liquidus isotherms in the tridymite and cristobalite (SiO_2) primary phase fields were measured, and the shape of the two liquid slags miscibility gap over cristobalite was determined for the "$CuO_{0.5}$"-"FeO"-SiO_2 and "FeO"-SiO_2 systems in equilibrium with metallic copper and iron respectively. The two liquid slags miscibility gap over cristobalite was found to be present at all Cu/(Cu+Fe) molar ratios in the slag. However, the difference in the SiO_2 concentration between the low- and high-SiO_2 slag was found to decrease as the Cu/(Cu+Fe) molar ratio in slag approached 0.5, with the minimal difference assessed to be approximately 4 mol.%. Additionally, new phase equilibria data obtained for the "FeO"-SiO_2 subsystem in equilibrium with metallic iron suggested that the monotectic temperature of the subsystem is 1673 ± 3 °C, and that the low- and high- SiO_2 slags contain 59.7 mol.% and 96.4 mol.% SiO_2 respectively. The present study was a continuation of previous investigations by the authors into the Cu-Fe-Si-O multicomponent slag system, aimed at providing information for improving the thermodynamic models for all phases in this system.

1. Introduction

The present work focused on the experimental phase equilibria of the "$CuO_{0.5}$"-"FeO"-SiO_2 and "FeO"-SiO_2 slag systems in equilibrium with metallic copper and iron respectively. These systems are important for the copper smelting, converting, and recycling processes, in addition to ironmaking. Although the range of temperatures studied (1400–1680 °C) were significantly higher than those encountered in nonferrous metallurgical processes, improvements to the thermodynamic model describing the liquid slag compositions (≥ 40 mol% SiO_2) corresponding to the temperatures studied will improve the accuracy of model predictions made when extrapolating the fundamental low order systems to the higher order systems associated with emerging nonferrous metallurgical processes. Therefore, the present study was an important part of the integrated thermodynamic modelling and experimental research program into the Pb-Zn-Cu-Fe-Si-Ca-Al-Mg-O system for nonferrous metal smelting and recycling industries.

Previous phase equilibria studies of the "$CuO_{0.5}$"-"FeO"-SiO_2 system were focused on the iron silicate slag in equilibrium with copper [1,2] or copper gold-alloys [3,4], with the Cu/(Cu+Fe) molar ratio in slag < 0.1. In those experiments, a synthetic iron-silica slag was equilibrated with copper or copper-gold alloy under an atmosphere with controlled P_{O_2}, then quenched, physically separated, and analysed using wet chemical or spectrophotometric techniques to measure the concentrations of the elements in the slag and the metal.

Ruddle et al. [1] used silica crucibles to study the iron silicate slag in equilibrium with metallic copper at 1200 °C, between the oxygen partial pressures of 10^{-6} atm (most oxidising) to metallic iron equilibrium (most reducing). Altman and Kellogg [3] used silica crucibles to investigate the iron silicate slag in equilibrium with copper-gold alloys, and extended the range of temperatures studied to 1200–1400 °C. Taylor and Jeffes [4] used a levitating drop technique to investigate the fully liquid iron silicate slag in equilibrium with copper-gold alloy at 1300–1450 °C, between the oxygen partial pressures of 10^{-5}–$10^{-9.3}$ atm. Oishi et al. [2] used a solid zirconia

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electrolyte galvanic cell to measure the oxygen activity of copper metal in equilibrium with the iron silicate slag between 1200 and 1300 °C, with silica crucibles used to contain the samples. After the attainment of equilibrium, they also quenched small quantities of the slag and metal phases for composition measurement.

All the studies discussed above focused on the effects of temperature and oxygen partial pressure variation on the copper solubility in the iron-silicate slag at equilibrium, or near equilibrium with tridymite (SiO_2).

More recently, Hidayat et al. [5] studied the "$CuO_{0.5}$"-"FeO"-SiO_2 system in equilibrium with air or metallic copper between 1100 and 1300 °C, using high temperature equilibration on suspended substrates followed by quenching and electron probe X-ray microanalysis (EPMA) to directly measure phase compositions. The Cu/(Cu+Fe) molar ratio in the slag investigated ranged from 0.05 to 1, which enabled the extents of the tridymite (SiO_2), cuprite (Cu_2O), delafossite ($CuFeO_2$), spinel ($(Cu,Fe)_3O_4$), and wüstite (FeO_{1+x}, x = 0.0238–0.0882) primary phase fields of the "$CuO_{0.5}$"-"FeO"-SiO_2 system to be determined for slags in the 5–40 mol.% SiO_2 concentration range. No phase equilibria data were found above 1450 °C for the "$CuO_{0.5}$"-"FeO"-SiO_2 system in equilibrium with metallic copper.

The experimental phase equilibria of the "$CuO_{0.5}$"-"FeO", "$CuO_{0.5}$"-SiO_2 and "FeO"-SiO_2 pseudo-binary systems in equilibrium with metal were investigated to different extents based on the limitations of the experimental techniques [6–18].

Phase equilibria data for the "$CuO_{0.5}$"-"FeO" system in equilibrium with metallic copper were found between the cuprite (Cu_2O)/delafossite ($CuFeO_2$) eutectic point (modelled to be 1128 °C by Hidayat et al. [19]) and 1400 °C. Yazawa and Eguchi [6] investigated the "$CuO_{0.5}$"-"FeO" system between 1150–1300 °C, and 0–45% mol "FeO_x" in slag. They applied differential thermal analysis (DTA) using platinum crucibles to investigate the cuprite (Cu_2O), delafossite ($CuFeO_2$) and spinel ($(Cu,Fe)_3O_4$) primary phase fields up to a maximum of 1275 °C. The compositions of the samples after the DTA experiments were not measured. Additionally, they equilibrated samples in platinum crucibles under an atmosphere of pure CO_2 or blended of N_2/O_2 to determine the course of oxygen isobars at isothermal sections (1200, 1250 and 1300 °C) of the Cu-Fe-O diagram. The quenched slag and the crucible from the equilibration experiments were analysed by chemical analysis and EPMA respectively. Takeda [7] investigated the "$CuO_{0.5}$"-"FeO" system between 1100–1400 °C, and 0–30 mol.% "FeO_x" in the slag. Samples were equilibrated in magnesia (MgO) crucibles to investigate the cuprite (Cu_2O), delafossite ($CuFeO_2$) and spinel ($(Cu,Fe)_3O_4$) primary phase fields. Ilyushechkin et al. [8], and Nikolic et al. [9] used the high temperature equilibration/rapid quenching/EPMA technique to investigate the cuprite (Cu_2O), delafossite ($CuFeO_2$) and spinel ($(Cu,Fe)_3O_4$) primary phase fields of the "$CuO_{0.5}$"-"FeO" system in equilibrium with metallic copper between 1150 and 1250 °C. Poor quenching of the slag in the "$CuO_{0.5}$"-"FeO" system from temperatures above 1250 °C limited the applicability of the quenching/EPMA technique. The liquidus temperature and compositions obtained using DTA by Yazawa and Eguchi [6] between 1100 and 1250 °C demonstrated good agreement with those measured during the later studies [8,9] using the quenching/EPMA technique. However, the results obtained by Takeda [7] showed poor agreement with the other results, likely because of the MgO contamination of the slag of up to 1 wt% and MgO dissolution in the spinel ($(Cu,Fe)_3O_4$) of up to 16 wt%.

Phase equilibria data for the "$CuO_{0.5}$"-SiO_2 system in equilibrium with metallic copper were found for the tridymite/cristobalite (SiO_2) primary phase fields between the cuprite/tridymite (SiO_2) eutectic point (modelled to be 1189 °C by Hidayat et al. [19]) and 1400 °C [10–13]. The results from the more recent studies that measured the slag composition using EPMA [10,11] showed good agreement in this temperature range with the earlier studies which used bulk wet chemical analysis [12,13]. The only experimental studies of the system found between 1400 °C and the two liquid slags miscibility gap over cristobalite (SiO_2) (modelled to be 1677 °C by Hidayat et al. [19]) were by Liu [14], and Shevchenko and Jak [15]. The results by Liu [14] were considered to be less accurate than the results from the later study by Shevchenko and Jak [15], as the former measured the slag composition using a zero probe diameter.

Phase equilibria studies on the "FeO"-SiO_2 slag system in equilibrium with metallic iron were recently reviewed by Hidayat et al. [20]. While the system was extensively studied between 1100 and 1400 °C, only 5 experimental data points were found at 1400 °C and above (consisting of results from Greig [16], Bowen and Schairer [17], and Turkdogan and Bills [18]). The results by Bowen and Schairer [17], and Turkdogan and Bills [18] were in agreement with respect to the shape of the tridymite and cristobalite (SiO_2) primary phase fields, but were inconsistent with the monotectic temperature and the compositions of the immiscible liquids reported by Greig [16].

The present study addressed the lack of experimental phase equilibria data at 1400 °C and above for the high-SiO_2 region of the "$CuO_{0.5}$"-"FeO"-SiO_2 and "FeO"-SiO_2 slag systems in equilibrium with metal and tridymite/cristobalite (SiO_2).

2. Experimental technique and procedure

2.1. Experimental and analysis techniques

The experimental/analytical techniques and equipment used for the present study were described in previous publication by the authors [21,22]. The starting mixtures were made from high purity powders of Cu_2O (≥99.9 wt% purity) and SiO_2 (≥99.9 wt% purity) supplied by Alfa Aesar, MA, USA, and Fe_2O_3 (≥99.9 wt% purity) and "FeO" (≥99.9 wt% purity) supplied by Sigma-Aldrich, MO, USA. The reagent "FeO" was assumed to be $FeO_{1.05}$ as the stable phase with the exact composition FeO does not exist at room temperature and pressure. The mixtures were planned with the goal of obtaining a solid oxide fraction of less than 50% by volume (preferably about 10%) to achieve equilibration and adequate outcomes from quenching (it was found that the solids served as heterogeneous nucleation centres, and that the minimum distance between them should exceed 10–20 µm to ensure that the slag was amorphous and uniform in composition, unaffected by solute ejection or dendrite growth). Cu powder (≥99.9 wt% purity) or Fe powder (≥99.9 wt% purity) supplied by Alfa Aesar, MA, USA were added in excess of the mass of the oxides by 10–20 wt% to ensure that the samples contained metallic copper or iron after the high temperature equilibration experiments.

Less than 0.5 g of mixture was used in each equilibration experiment. The mixtures were pressed into pellets using a tool steel die before being inserted into quartz (SiO_2) tubes. The tubes were then sealed into ampoules under vacuum (~0.001 atm) to limit the evaporation of copper and copper oxides from the mixture at high temperatures, and to limit the oxidation of the mixture by residual oxygen in the furnace atmosphere.

The high temperature equilibration experiments were performed in the vertical tube resistance heated furnace. The reaction tube was made from impervious recrystallised alumina (Al_2O_3) with 30-mm inner diameter (supplied by the Xiamen Wintrustek Advanced Materials Company, Fujian, China). The furnace controller thermocouple was periodically calibrated using a B-type thermocouple, which in turn was calibrated against a standard thermocouple supplied by the National Measurement Institute of Australia, NSW, Australia. The overall temperature accuracies of the experiments were estimated to be ± 3 K.

The sealed ampoules containing the pelletised mixtures were suspended in the hot zone of the furnace by Kanthal-D wire (Fe-Al-

Cr alloy, 0.7 or 1-mm diameter, melting point 1500 °C). A 10–25 cm long piece of platinum wire was used to extend the support for the equilibration experiments at ≥ 1500 °C so that the Kanthal wire was not in the hot zone. Argon gas (99.999% purity, supplied by Coregas, NSW, Australia) was passed through the furnace during equilibration to prevent sample oxidation.

The samples were first pre-melted for 10–20 min at 5–20 °C above the final equilibration temperature to form a homogenous liquid slag phase, followed by equilibration at the final target temperature. The samples were quenched into 30% $CaCl_2$ brine at − 20 °C at the end of the equilibration process, washed using water and ethanol, dried in a convection oven at 40 °C, then mounted in epoxy resin and cross-sectioned using conventional metallographic polishing techniques.

The cross-sectioned samples were examined under the optical microscope, carbon-coated, and the compositions of the phases were then measured using an Electron Probe X-ray Microanalyser (JEOL 8200L EPMA; Japan Electron Optics Ltd., Tokyo, Japan) with the wavelength-dispersive spectrometer. The microanalyser was operated with 15 kV accelerating voltage and 20 nA probe current. The Duncumb-Philibert atomic number, absorption, and fluorescence (ZAF) correction supplied by JEOL [23–25] was applied. Copper (Cu), haematite (Fe_2O_3) and quartz (SiO_2) (supplied by the Charles M. Taylor Co., Stanford, CA), and cuprite (Cu_2O, prepared by oxidising high purity copper foil supplied by BDH Chemicals Ltd, Poole, United Kingdom) were used as the EPMA standards. The concentrations of the metal cations were measured with EPMA; no data on the oxidation states of the metal cations were obtained. For presentation purposes, the copper oxide concentration was then re-calculated as "$CuO_{0.5}$", and the iron oxide concentration was re-calculated as "$FeO_{1.5}$" for the "$CuO_{0.5}$"-"FeO_x"-SiO_2 pseudo-ternary projection, and as "FeO" for the "FeO"-SiO_2 pseudo-binary projection. The molar ratios of the oxides were used in the representation of the data.

For accurate, repeatable, and objective measurement of the average composition of the liquid slag phase by EPMA, an approach similar to that described by Nikolic et al. [26] was used. The slag composition was taken to be the average of 20 points in the best quenched area (usually near the slag/crucible interface), and the results were rejected if the standard deviation in the measured concentration of any metallic cation in the slag exceeded 1 mol.%.

The ability to quench the liquid slag phase to ambient temperature without the onset of crystallisation was found to depend on the composition of the slag and the equilibration temperature. Tridymite or cristobalite (SiO_2) microcrystals (< 1 µm, rounded in appearance) were more likely to precipitate upon quenching from the low-SiO_2 slags when compared to the high-SiO_2 slags (liquidus temperature > 1600 °C). The probability of microcrystals formation on quenching also increased with the increasing Cu/(Cu+Fe) molar ratio in the liquid slag. Additionally, most samples contained fine copper metallic droplets (< 1 µm), possibly formed on quenching via the following reaction:

$$CuO_{0.5(slag)} + FeO_{(slag)} \rightarrow Cu_{(metal)} + FeO_{1.5(slag)} \quad (1)$$

Probe diameters of 20–50 µm were therefore used to analyse the liquid slag phase of the samples. Poorly quenched regions (due to solute ejection from nearby tridymite/cristobalite (SiO_2) crystals, or copper oxide ejection from cooling metallic copper droplets) were avoided.

2.1.1. Custom correction of the standard JEOL ZAF correction of measurements made using K_α X-rays

The accuracies of the EPMA compositional measurements made using K_α X-rays were improved by applying secondary corrections to the concentrations obtained after the standard ZAF correction supplied by JEOL, following an approach similar to that described in previous publications by the authors [27]. The compositions of stoichiometric dehydrated dioptase ($CuSiO_3$, prepared by dehydrating single crystals of natural dioptase, $CuSiO_3{\cdot}H_2O$, procured from the Altyn-Tyube dioptase deposit, Karaganda, Kazakhstan at 600–650 °C for 2 h) and fayalite (Fe_2SiO_4, obtained during equilibration experiments performed using the same experimental methodology as the present study) were measured using EPMA in the previous studies [27,28] using the same probe accelerating voltages and currents, and ZAF corrections supplied by JEOL [23–25] as the present study. It was found that after the application of the standard JEOL ZAF corrections, the concentration of SiO_2 in dioptase ($CuSiO_3$) and in fayalite (Fe_2SiO_4) were overestimated by 1.8 mol.% [27] and 1.0 mol.% [28] respectively. Based on these reference points, the following correction formulae were developed:

$$x_{CuO_{0.5}}^{Corrected} = x_{CuO_{0.5}}(1 + 7.2 \times 10^{-2} x_{SiO_2}), \quad (2)$$

$$x_{FeO_x}^{Corrected} = x_{FeO_x}(1 + 6.73 \times 10^{-2} x_{SiO_2}(1 - x_{SiO_2})), \quad (3)$$

$$x_{SiO_2}^{Corrected} = 1 - \sum_{i \neq SiO_2} x_i^{Corrected}, \quad (4)$$

where x_i refers to the normalised molar fractions of each metal(loid) oxide in a phase after the initial standard JEOL ZAF corrections, and $x_i^{Corrected}$ refers to the corrected, normalised molar fraction of each metal(loid) oxide in a phase.

2.1.2. Custom correction of the standard JEOL ZAF correction of measurements made using L_α X-rays

Both characteristic X-rays and background radiation (Bremsstrahlung) are known to excite areas of the sample outside of the electron beam spot during EPMA. This results in secondary X-rays being generated in the indirectly excited areas, and the effect was termed secondary X-ray fluorescence [29].

The concentrations of impurities in various phases may be overestimated by the EPMA measurements due to the generation of secondary fluorescent X-rays in the adjacent phases. Previous publications by the authors [15,30,31] demonstrated that the Cu, Fe and Zn concentrations in the SiO_2 solids were overestimated by the EPMA measurements; when measured using their K_α X-rays, non-zero Cu, Fe and Zn apparent concentrations were reported in 10–100 µm pure SiO_2 particles embedded within unreacted pellets of Cu [15], Fe_2O_3 [30,31], or ZnO [30,31].

The secondary fluorescence effect was detected for the X-rays with energies > 5 keV, as these X-rays have long (> 100 µm) penetration range in the SiO_2 solids and materials of comparable density, such as the silicate slags. X-rays with low energies, such as the L_α X-rays of Cu (0.93 keV) and Fe (0.705 keV), cannot penetrate far into the SiO_2 solids or materials of comparable density, and consequently do not result in the secondary fluorescent X-rays from adjacent

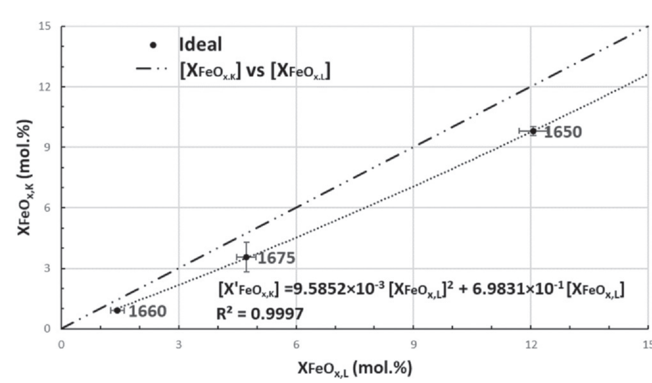

Fig. 1. "FeO_x" concentration measured in the slag using the K_α X-ray, $XFeO_{x,k}$ (mol.%) vs "FeO_x" concentration measured in slag using the L_α X-ray, $XFeO_{x,L}$ (mol.%). Equilibration temperatures are in °C, and the error bar values represent one standard deviation of measurement uncertainty.

Table 1
Typical lower detection limits (reported to 2 significant figures) of Cu and Fe measured in SiO2 using EPMA, as calculated using the 3SDBG criterion suggested by Love et al. [35].

Element	X-ray	WDS Crystal	Standard	BG Measurement Time (s)	Probe Current (nA)	Background Intensity in SiO_2 (CPS/nA)	Standard Peak Intensity (CPS/nA)	Standard K-Factor	ZAF in SiO_2	Typical LDL (wt%)
Cu	Lα	Thallium Acid Phthalate (TAP)	Cu metal	5	20	1.0	170	0.12	1.6[a]	0.05
Fe	Lα	Artificial Layered Dispersive Element (LDE1)	Haematite (Fe_2O_3)	5	20	2.2	42	0.13	1.1[a]	0.22

[a] Value reported by the probe software for measurements when trace quantities (< 0.05%) of Cu or Fe were detected.

phases. The concentrations of "$CuO_{0.5}$" and "FeO_x" in tridymite and cristobalite (SiO_2) from several samples were measured in the present study using their L_α X-rays to accurately determine their solubilities [29,32].

Preliminary measurements indicated that the "FeO_x" concentration measured using its L_α X-ray systematically differed from the "FeO_x" concentration measured using its K_α X-ray, indicating significant uncertainties in the corresponding standard JEOL ZAF corrections. To overcome this issue, the concentrations of "FeO_x" in several well quenched samples with 1–12 mol.% "FeO_x" concentration were measured using both the K_α and L_α X-rays of "FeO_x" to:

a) Confirm the lower detection limit for the "FeO_x" concentrations measured using the L_α X-ray; and
b) Determine an empirical relation between the "FeO_x" concentrations measured using the K_α and L_α X-rays.

The relationship between the "FeO_x" molar concentration in the slag measured using the K_α and L_α X-rays is shown in Fig. 1. Based on the differences in the measured concentrations, the following correlation was derived:

$$[X'_{FeO_{x,K}}] = 9.5852 \times 10^{-3}[X_{FeO_{x,L}}]^2 + 6.9831 \times 10^{-1}[X_{FeO_{x,L}}], \quad (5)$$

where $X'FeO_{x,K}$ is the normalised "FeO" molar concentration measured using the K_α X-ray after standard JEOL correction, and $XFeO_{x,L}$ is the normalised "FeO" molar concentration measured using the L_α X-ray after standard JEOL correction.

The following correlation (reported by Khartcyzov et al. [33]) was used to correct the "$CuO_{0.5}$" concentration measured using the L_α X-ray, $XCuO_{0.5,L}$ (mol.%):

$$[X'_{CuO_{0.5,K}}] = -1.2719 \times 10^{-3}[X_{CuO_{0.5,L}}]^2 + 1.1261[X_{CuO_{0.5,L}}], \quad (6)$$

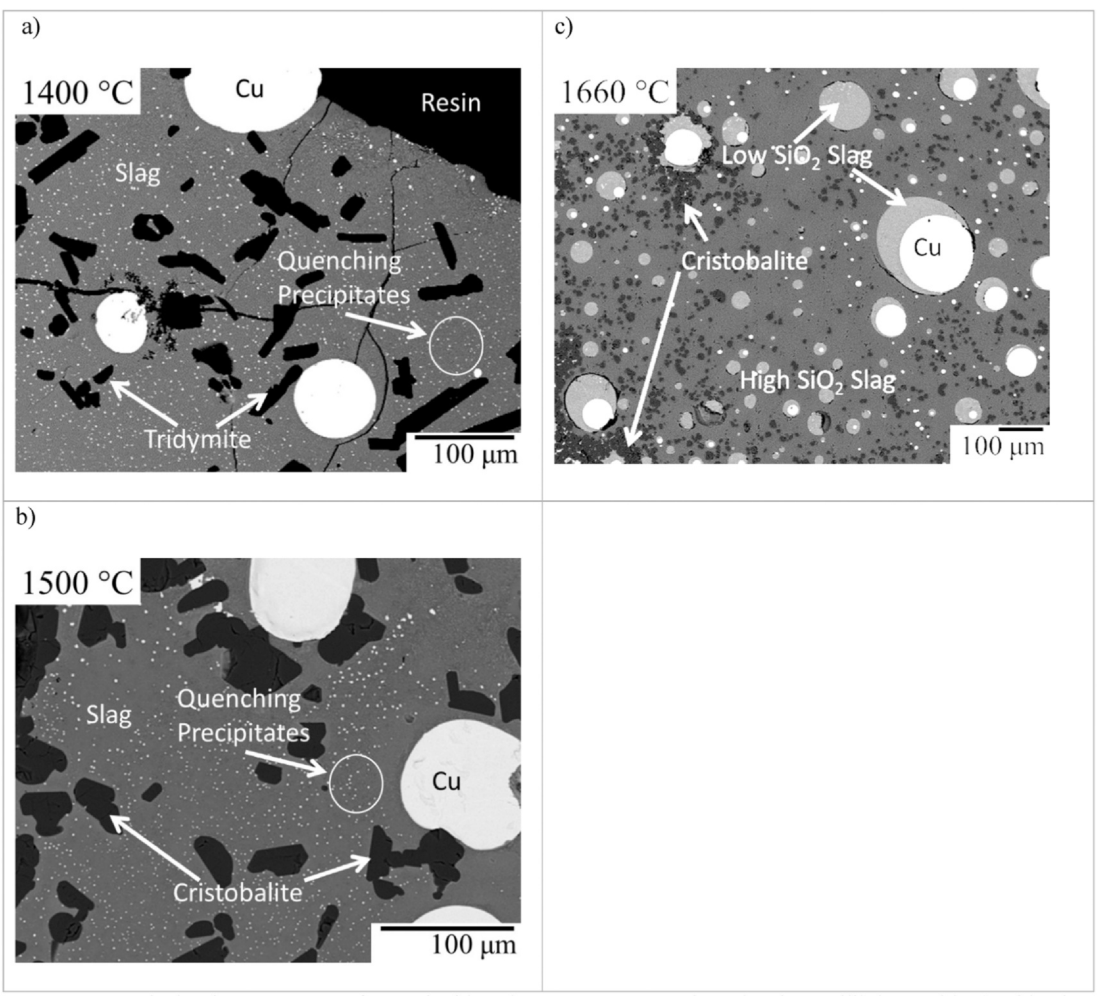

Fig. 2. Typical Microstructures of Cu Liquid and "$CuO_{0.5}$"-"FeO"-SiO_2 slag in equilibrium with a) tridymite (SiO_2); b) cristobalite (SiO_2); and c) a second liquid slag and cristobalite (SiO_2). Quenching precipitates are metallic copper droplets formed on quenching by the reaction described in Eq. (1).

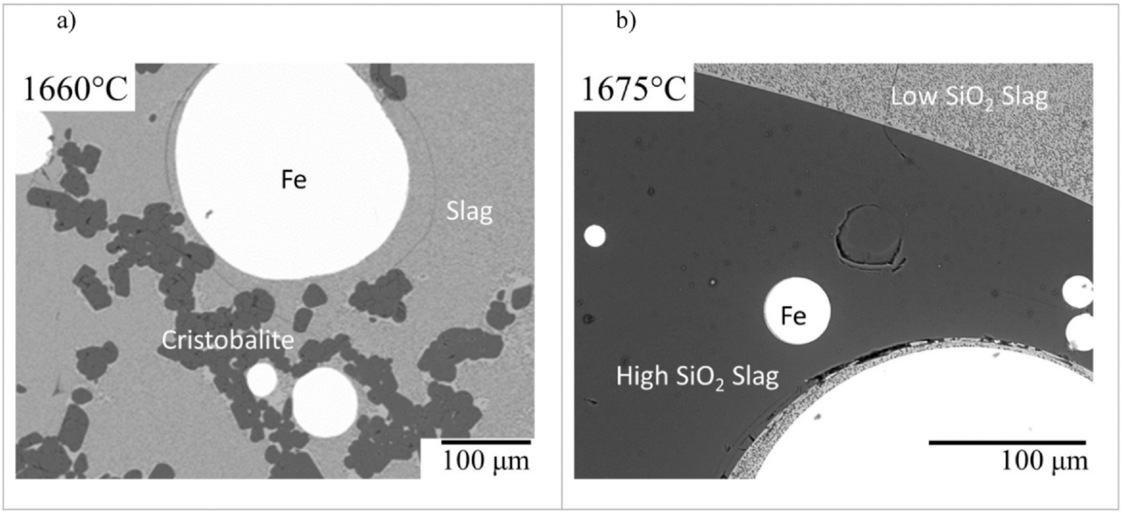

Fig. 3. Typical Microstructures of Fe metal and "FeO"-SiO_2 slag in equilibrium with a) with cristobalite (SiO_2); and b) a second liquid slag.

where $X'CuO_{0.5,K}$ is the normalised "$CuO_{0.5}$" molar concentration measured using the K_α X-ray after standard JEOL correction, and $XCuO_{0.5,L}$ is the normalised "$CuO_{0.5}$" molar concentration measured using the L_α X-ray after standard JEOL correction.

Eqs. (5) and (6) were applied to normalised "FeO_x" and "$CuO_{0.5}$" concentrations measured using their L_α X-rays after JEOL ZAF corrections to calculate the pseudo K_α concentrations. Eqs. (2)–(4) were then applied to the pseudo "FeO_x" and "$CuO_{0.5}$" K_α concentrations to further correct the measured concentrations of these elements in tridymite/cristobalite (SiO_2).

2.1.3. Lower detection limits of trace elements in tridymite/cristobalite (SiO_2) during electron probe microanalysis

The presence of background radiation, and beam current and spectrometer instability during EPMA imposes a lower limit on the minimum concentration of an element that can be reliably detected [34]. Quantitatively, the single point lower detection limit is usually defined with relation to the variability in the background count intensity on the sample and the peak count intensity on a well-defined standard. Love et al. [35] recommended that the lower detection limit of EPMA should correspond to a peak intensity equal to three standard deviations of the background count above the background intensity, as this corresponds to a confidence level of > 99% that an element is truly present in the sample. This is expressed by the formula:

$$LDL_{3SDBG} = ZAF_B \times 3\frac{\sqrt{N_B}}{t_B C_B} \times \frac{K_s}{I_S} \quad (7)$$

Where ZAF_B is the ZAF correction factor of the measured element in the sample matrix, N_B is the total number counts on the background of the sample, t_B is the background count time (in seconds), C_B is the current intensity used for background measurements (in nA), I_s the peak intensity in the standard (in cps/nA) and K_S is the K-factor of the standard (defined as the concentration of the element in the standard divided by the ZAF correction for the element in the standard).

To quantify the detection limits of Cu and Fe measured in tridymite and cristobalite (SiO_2), the pure SiO_2 standard was measured at the background positions used in the Cu and Fe Lα X-rays measurements. The lower detection limits calculated from this are summarised in Table 1.

2.2. Confirmation of the achievement of equilibrium in the experiments

The "4-point test" [21,22] was applied to ensure the achievement of equilibrium in the experiments. This consisted of: a) the variation of the equilibration time; 2) the assessment of the compositional homogeneity of the phases by EPMA; 3) approaching the final equilibrium point from different starting compositions; and 4) the consideration of reactions specific to the system that can affect the

Table 2
Summary of apparent "$CuO_{0.5}$" and "FeO_x" concentrations in tridymite, cristobalite or unreacted amorphous SiO_2, as measured in previous studies [5,10,11,14,15,38,39] using EPMA and Kα X-rays.

		mol% $CuO_{0.5}$		mol% $FeO_{1.5}$	
Study	SiO_2 Polymorph	min	max	min	max
Liu [14]	Cristobalite**	1.01	1.43	N/A	
Hidayat et al. [5]	Tridymite*	0.08	1.18	0.08	0.75
Hidayat et al. [10]	Tridymite**	0.76	1.26	*N/A*	
Xia et al. [11]	Tridymite**	0.71	1.35	*N/A*	
Fallah-Mehrjardi et al. [38]	Tridymite*	0.00	0.62	0.23	1.53
Hidayat et al. [39]	Tridymite*	0.00		0.56	
Shevchenko and Jak [15]	Amorphous, in unreacted pellet of SiO_2+ Cu	1.05		*N/A*	
Shevchenko and Jak [28]	Tridymite***	*N/A*		0.00	0.74
Shevchenko and Jak [30]	Tridymite***	*N/A*		0.02	0.12
Shevchenko and Jak [30]	Cristobalite***	*N/A*		0.05	0.24
Present Study	**Tridymite**	**0.23**	**0.63**	**0.43**	**0.61**
	Cristobalite	**0.09**	**1.30**	**0.05**	**0.71**

*Study of the "$CuO_{0.5}$"-"FeO_x"-SiO_2 slag system; the measured concentrations of Cu and Fe measured in tridymite/cristobalite (SiO_2) correlated directly with their respective concentrations in the slag **Study of the "$CuO_{0.5}$"- SiO_2 slag system **Study of the PbO-"FeO_x"-SiO_2 system; results are from samples measured as containing 0% PbO in tridymite or cristobalite (SiO_2).

Table 3
Experimental results for the "$CuO_{0.5}$"-"FeO"-SiO_2 system in equilibrium with Cu Metal. Measured elemental concentrations in the copper liquid have been shaded in grey.

Pre-melt °C	Final T, °C	Time, h	Phase	Measured composition, wt. %				Normalised and corrected composition, mol. %		
				SiO_2 [Si]	$FeO_{1.5}$ [Fe]	$CuO_{0.5}$ [Cu]	Total	SiO_2 [Si]	$FeO_{1.5}$ [Fe]	$CuO_{0.5}$ [Cu]
Tridymite (SiO_2) Liquidus										
1420	1400	0.3 + 4	Slag	22.9±0.30	12.6±0.16	65.5±0.52	101.0±0.42	24.9±0.42	11.0±0.14	64.0±0.49
			Tridymite	99.97±0.53	0.00±0.00^L	0.05±0.01^L	100.0±0.54	99.96±0.01	0.00±0.00^L	0.04±0.01^L
			Cu liquid**	0.02	0.05	100.00	100.1	0.04	0.05	99.91
1420	1400	0.3 + 4	Slag	31.1±0.88	30.5±0.66	40.1±1.02	101.7±0.69	34.1±0.96	26.5±0.63	39.3±0.86
			Tridymite**	99.88	0.00^L	0.00^L	99.9	100.00	0.00^L	0.00^L
			Cu liquid	0.00	0.12	99.63	99.7	0.00±0.00	0.03±0.04	99.97±0.04
1420	1400	0.3 + 4	Slag	31.8±0.63	34.6±0.40	36.4±0.64	102.8±0.59	34.7±0.60	29.9±0.47	35.4±0.51
			Tridymite**	101.74	0.00^L	0.05^L	101.8	99.96	0.00^L	0.04^L
			Cu liquid**	0.00	0.04	99.87	99.9	0.00	0.05	99.95
1420	1400	0.3 + 4	Slag	33.3±0.61	37.1±0.61	33.1±1.13	103.5±0.67	36.1±0.65	31.9±0.61	32.1±0.97
			Tridymite**	100.14	0.00^L	0.00^L	100.1	100	0.00^L	0.00^L
			Cu liquid	0.00±0.00	0.03±0.05	100.24±0.04	100.3±0.00	0.00±0.00	0.03±0.04	99.97±0.04
1420	1400	0.2 + 4	Slag	33.7±0.54	38.9±0.51	30.9±0.73	103.5±0.81	36.6±0.49	33.5±0.37	29.9±0.66
			Tridymite*	96.5±1.48	0.79±0.01	0.69±0.00	97.0±99.07	>98.8±0.01	<0.61±0.00	<0.63±0.01
			Cu liquid	0.00±0.00	0.19±0.05	98.7±0.11	98.9±0.07	0.00±0.00	0.17±0.04	99.83±0.04
1420	1400	0.3 + 4	Slag	36.3±1.07	49.0±0.88	18.1±0.87	103.4±0.74	40.0±0.95	42.3±1.00	17.7±0.90
			Tridymite*,**	101.32	0.58	0.27	102.2	>99.34	<0.43	<0.23
			Cu liquid**	0.00	0.05	99.89	99.9	0.00	0.05	99.95
Cristobalite (SiO_2) Liquidus										
1520	1500	0.3 + 3	Slag	32.4±1.04	10.2±0.36	59.4±0.99	102.0±1.18	34.5±0.91	8.6±0.30	56.9±0.90
			Cristobalite	100.54±0.57	0.00±0.00^L	0.00±0.00^L	100.5±0.57	100.00±0.00	0.00±0.00^L	0.00±0.00^L
			Cu liquid	0.02±0.01	0.01±0.01	98.5±0.68	98.5±0.68	0.03±0.03	0.00±0.01	99.97±0.03
1520	1500	0.3 + 3	Slag	37.4±0.72	20.6±0.52	43.3±0.90	101.4±0.69	40.4±0.70	17.6±0.50	42.0±0.72
			Cristobalite	101.05±0.18	0.00±0.00	0.00±0.01	101.1±0.17	100.00±0.00	0.00±0.00^L	0.00±0.00^L
			Cu liquid**	0.00	0.19	98.4	98.6	0.00	0.17	99.83
1520	1500	0.3 + 0.5	Slag	38.5±0.65	32.7±0.39	27.3±0.99	98.5±0.38	43.4±0.79	29.1±0.41	27.5±0.91
			Cristobalite	100.31±0.60	0.00±0.00^L	0.00±0.00^L	100.3±0.60	100.00±0.00	0.00±0.00^L	0.00±0.00^L
			Cu liquid	0.00±0.00	0.47±0.28	99.62±0.00	100.1±0.27	0.00±0.01	0.54±0.32	99.46±0.31
1520	1500	0.3 + 3	Slag	44.4±0.83	48.8±0.87	11.9±1.12	105.1±0.86	47.7±0.84	41.0±0.86	11.3±0.99
			Cristobalite*,**	98.1	0.93	0.24	99.3	>99.07	<0.71	<0.22
			Cu liquid**	0.00	0.70	100.48	101.2	0.00	0.62	99.38
1610	1600	0.3 + 1.5	Slag	40.3±0.92	9.0±0.46	52.0±0.71	101.2±0.60	42.8±0.80	7.6±0.39	49.7±0.74
			Cristobalite	100.73±0.52	0.00±0.00^L	0.00±0.00^L	100.7±0.52	100.00±0.00	0.00±0.00^L	0.00±0.00^L
			Cu liquid	0.00±0.00	0.01±0.01	98.3±0.41	98.3±0.42	0.00±0.00	0.03±0.00	99.97±0.00
1620	1600	0.3 + 2	Slag	46.4±0.74	15.1±0.30	39.2±0.54	100.7±0.88	49.6±0.49	12.7±0.19	37.7±0.55
			Cristobalite**	100.32	0.00^L	0.00^L	100.3	100.00	0.00^L	0.00^L
			Cu liquid	0.00±0.00	0.04±0.02	97.5±0.29	97.5±0.28	0.00±0.00	0.04±0.02	99.96±0.02
1620	1600	0.3 + 2	Slag	51.4±0.56	21.3±0.35	29.6±0.57	102.3±0.62	54.3±0.46	17.7±0.29	28.0±0.49
			Cristobalite**	99.97	0.00^L	0.00^L	100.0	100.00	0.00^L	0.00^L
			Cu liquid	0.00±0.00	0.17±0.05	98.1±0.33	98.3±0.38	0.00±0.00	0.15±0.04	99.85±0.04
1620	1600	0.3 + 2	Slag	50.9±0.67	25.3±0.63	25.7±1.01	101.9±0.54	54.5±0.83	21.1±0.57	24.5±0.87
			Cristobalite*	99.86±2.90	0.59±0.12	0.90±0.31	101.3±2.47	>98.8±0.38	<0.44±0.10	<0.81±0.28
			Cu liquid**	0.00	0.13	97.0	97.2	0.00	0.12	99.88
1620	1600	0.3 + 2	Slag	50.7±1.51	39.5±0.56	9.7±0.53	100.0±1.53	56.3±0.91	34.1±0.54	9.6±0.56
			Cristobalite*	103.24±0.35	0.79±0.04	0.33±0.02	104.4±0.36	>99.80±0.08	<0.11±0.03	0.09±0.04
			Cu liquid	0.00±0.00	0.27±0.07	100.66±0.21	100.9±0.28	0.00±0.00	0.56±0.41	99.44±0.41
1610	1600	0.167 + 4	Slag	54.0±0.56	43.4±0.68	7.8±0.87	105.2±0.44	57.0±0.60	35.6±0.57	7.4±0.78
			Cristobalite**	98.8	0.00^L	0.04^L	98.8	99.97	0.00^L	0.03^L
			Cu liquid	0.00±0.00	0.03±0.02	98.9±0.34	98.9±0.34	0.00±0.00	0.03±0.02	99.97±0.02

(continued on next page)

Table 3 (*continued*)

Pre-melt °C	Final T, °C	Time, h	Phase	Measured composition, wt. %				Normalised and corrected composition, mol. %		
				SiO_2 [Si]	$FeO_{1.5}$ [Fe]	$CuO_{0.5}$ [Cu]	Total	SiO_2 [Si]	$FeO_{1.5}$ [Fe]	$CuO_{0.5}$ [Cu]
1610	1600	0.083 + 0.5	Slag	52.2±0.69	50.3±0.95	3.8±1.00	106.3±0.48	55.1±0.72	41.1±0.79	3.8±0.85
			Cristobalite**	100.00	0.00^L	0.00^L	100.0	100.00	0.00^L	0.00^L
			Cu liquid	0.00±0.00	0.09±0.03	99.89±0.00	100.0±0.03	0.00±0.00	0.08±0.03	99.92±0.03
1610	1600	0.3 + 5	Slag	50.7±0.35	50.9±0.41	4.1±0.39	105.7±0.34	54.0±0.32	42.2±0.34	3.9±0.36
			Cristobalite	101.46±0.88	$0.00±0.00^L$	$0.00±0.00^L$	101.5±0.88	100.00±0.00	$0.00±0.00^L$	$0.00±0.00^L$
			Cu liquid	0.02±0.04	0.42±0.11	99.03±0.93	99.5±0.96	0.02±0.04	0.38±0.10	99.60±0.14
1645	1640	0.3 + 1.5	Slag	49.2±0.67	9.8±0.56	44.0±0.85	103.0±0.57	51.0±0.63	8.1±0.46	40.9±0.71
			Cristobalite*	100.86±0.45	0.27±0.00	1.3±0.42	102.4±0.03	>98.7±0.35	<0.20±0.00	<1.1±0.35
			Cu liquid	0.00±0.00	0.11±0.05	98.7±0.73	98.9±0.69	0.00±0.00	0.10±0.04	99.90±0.04
1650	1640	0.3 + 1.5	Slag	51.6±0.97	11.2±0.33	38.4±0.79	101.2±0.91	54.4±0.70	9.2±0.25	36.4±0.80
			Cristobalite*,**	97.8	0.21	1.4	99.4	>98.5	<0.16	<1.3
			Cu liquid	0.00±0.00	0.15±0.15	99.11±0.68	99.3±0.83	0.00±0.00	0.13±0.13	99.87±0.13
1650	1640	0.3 + 1.5	Slag	53.1±0.69	11.1±0.27	37.3±0.82	101.4±0.61	55.8±0.71	9.1±0.20	35.2±0.70
			Cristobalite*,**	98.7	0.25	1.4	100.3	>98.6	<0.19	<1.2
			Cu liquid	0.47±0.03	0.00±0.00	97.7±0.96	98.2±0.92	0.52±0.05	0.00±0.00	99.48±0.05
1650	1640	0.3 + 1.5	Slag	63.1±0.94	22.7±0.50	12.5±0.88	98.2±0.74	68.9±0.87	19.1±0.38	12.0±0.89
			Cristobalite*,**	94.0	0.53	0.42	95.0	>99.18	<0.42	<0.40
			Cu liquid	0.00±0.00	0.29±0.03	98.8±0.03	99.1±0.06	0.00±0.00	0.26±0.02	99.74±0.02
1660	1640	0.3 + 1.5	Slag	59.4±0.55	36.0±0.60	8.1±0.78	103.5±0.60	62.9±0.47	29.5±0.51	7.6±0.69
			Cristobalite*,**	102.19	0.86	0.52	103.6	>98.9	<0.63	<0.45
			Cu liquid***					N/A	N/A	~100
1654	1649	0.15 + 4	Slag	84.9±0.39	8.2±0.23	7.3±0.25	100.4±0.41	86.9±0.32	6.4±0.18	6.7±0.21
			Cristobalite*	100.68±0.13	0.26±0.04	0.56±0.01	101.5±0.17	>99.31±0.03	<0.19±0.03	<0.49±0.01
			Cu liquid	0.00±0.00	0.02±0.02	98.6±0.29	98.7±0.29	0.00±0.00	0.02±0.02	99.98±0.02
1658	1653	0.15 + 8	Slag	87.7±0.57	5.6±0.15	6.8±0.24	100.1±0.50	89.2±0.16	4.4±0.06	6.4±0.13
			Cristobalite*	100.07±1.41	0.19±0.05	0.57±0.04	100.8±1.33	>99.35±0.08	<0.14±0.04	<0.51±0.04
			Cu liquid	0.00±0.00	0.30±0.24	99.97±0.75	100.3±0.51	0.00±0.00	0.27±0.22	99.73±0.22
2 Immiscible Liquid Slags + Cristobalite (SiO_2)										
1668	1663	0.15 + 4	Low SiO_2 slag	52.3±0.46	7.7±0.23	42.6±0.51	102.6±0.29	54.1±0.41	6.3±0.18	39.6±0.48
			High SiO_2 slag	90.4±0.70	1.7±0.12	7.8±0.65	99.9±0.48	91.6±0.63	1.3±0.09	7.1±0.55
			Cristobalite*	100.62±0.53	0.11±0.03	0.58±0.05	101.3±0.51	>99.41±0.02	0.08±0.02	<0.51±0.04
			Cu liquid**	0.00	0.05	101.89	101.9	0.00	0.05	99.95
1661	1656	0.15 + 4	Low SiO_2 slag	61.5±0.44	9.1±0.38	31.4±0.55	101.9±0.63	63.6±0.43	7.3±0.30	29.1±0.38
			High SiO_2 slag	86.3±0.87	3.2±0.23	10.9±0.65	100.3±0.35	87.6±0.72	2.5±0.18	9.9±0.56
			Cristobalite*	100.03±0.65	0.07±0.06	0.47±0.06	100.6±0.67	>99.53±0.09	0.05±0.04	<0.42±0.05
			Cu liquid	0.00±0.00	0.08±0.02	101.64±0.99	101.7±0.99	0.00±0.00	0.07±0.01	99.93±0.01
1653	1648	0.15 + 4	Low SiO_2 slag	72.3±1.08	14.3±0.57	15.2±0.61	101.8±0.34	75.5±0.34	11.0±0.22	13.5±0.21
			High SiO_2 slag	78.3±0.34	10.9±0.21	12.0±0.12	101.2±0.32	80.4±0.23	8.6±0.17	11.0±0.09
			Cristobalite	100.00±0.33	0.00±0.00	0.01±0.01	100.0±0.32	100.00±0.00	$0.00±0.00^L$	$0.00±0.00^L$
			Cu liquid	0.00±0.00	0.12±0.01	97.5±1.21	97.6±1.20	0.00±0.00	0.11±0.01	99.89±0.01
1655	1650	0.15 + 4	Low SiO_2 slag	69.1±1.05	27.1±0.94	5.9±0.84	102.1±0.55	72.7±0.71	21.9±0.84	5.5±0.74
			High SiO_2 slag	82.6±0.81	12.6±0.27	5.9±0.21	101.1±0.81	84.7±0.38	9.9±0.22	5.4±0.19
			Cristobalite*	99.0±1.72	0.58±0.13	0.46±0.18	100.0±1.49	>99.14±0.20	<0.44±0.10	<0.42±0.15
			Cu liquid	0.00±0.00	0.21±0.20	99.20±0.55	99.4±0.70	0.00±0.00	0.18±0.18	99.82±0.18
1665	1660	0.3 + 1.5	Low SiO_2 slag	61.6±0.35	38.6±1.20	3.6±1.17	103.7±0.60	65.2±0.24	31.5±1.00	3.3±1.04
			High SiO_2 slag	90.4±1.24	8.3±0.36	2.1±0.14	100.9±1.26	91.7±0.36	6.4±0.28	1.9±0.12
			Cristobalite***					~100	N/A	N/A
			Cu liquid	0.00±0.00	0.02±0.00	100.97±0.05	101.0±0.05	0.00±0.00	0.02±0.00	99.98±0.00

*Fe and Cu concentration in tridymite/cristobalite could not be measured accurately due to the secondary X-ray fluorescence effect; the real concentration was expected to be lower than 0.05 mol%. **Only one successful measurement was made ***Estimated composition. The phase was seen in the micrographs, but not successfully measured. [L]The concentration of the element was measured using its L_α X-ray.

achievement of equilibrium or reduce the accuracy of the results. Several dedicated series of experiments were carried out for these purposes, resulting in the experimental design features described below.

3. Results and discussion

Typical quenched microstructures of the "$CuO_{0.5}$"-"FeO"-SiO_2 slag in equilibrium with metallic copper and the "FeO"-SiO_2 slag in equilibrium with metallic iron are shown in Figs. 2 and 3 respectively. For both systems, the tridymite/cristobalite (SiO_2) transition was not determined directly and was accepted to occur at 1465 °C in accordance with the FactSage thermodynamic database [36]. On micrographs, tridymite crystals were euhedral and plate-like in appearance, while cristobalite crystals were subhedral and equiaxed in appearance.

The observed phase assemblages and measured phase compositions are shown in Tables 3 and Table 4. The projections of these points onto the "$CuO_{0.5}$"-"FeO"-SiO_2 pseudo-ternary and the "FeO"-SiO_2 pseudo-binary diagrams are shown in Figs. 4 and 5 respectively.

Table 4
Experimental results for the "FeO"-SiO_2 system in equilibrium with metallic iron.

Pre-melt T, °C	Final T, °C	Time, h	Phase	Measured composition, wt%			Normalised and corrected Composition, mol. %	
				SiO_2	FeO	Total	SiO_2	FeO
Tridymite (SiO_2)								
1420	1400	0.2 + 2	Slag	42.10 ± 0.19	56.80 ± 0.28	98.9 ± 0.32	45.97 ± 0.16	54.03 ± 0.16
			Tridymite	101.71 ± 0.44	0.00 ± 0.00[a]	101.7 ± 0.44	100 ± 0.00	0.00 ± 0.00[a]
1470	1450	0.2 + 2	Slag	43.48 ± 0.16	56.52 ± 0.16	99.0 ± 1.01	47.03 ± 0.16	52.97 ± 0.16
			Tridymite	99.99 ± 1.13	0.00 ± 0.00[a]	100.0 ± 1.13	100 ± 0.00	0.00 ± 0.00[a]
Cristobalite (SiO_2)								
1520	1500	0.2 + 1.5	Slag	44.93 ± 0.19	55.07 ± 0.19	100.2 ± 0.33	48.53 ± 0.20	51.47 ± 0.20
			Cristobalite	101.45 ± 0.22	0.00 ± 0.00[a]	101.5 ± 0.22	100 ± 0.00	0.00 ± 0.00[a]
1560	1550	0.2 + 1	Slag	46.76 ± 0.15	53.24 ± 0.15	100.4 ± 0.42	50.41 ± 0.15	49.59 ± 0.15
			Cristobalite	101.76 ± 0.70	0.00 ± 0.00[a]	101.8 ± 0.70	100 ± 0.00	0.00 ± 0.00[a]
no	1600	1	Slag	49.74 ± 0.15	50.26 ± 0.15	100.8 ± 0.24	53.43 ± 0.15	46.57 ± 0.15
			Cristobalite	102.03 ± 0.25	0.00 ± 0.00[a]	102.0 ± 0.25	100 ± 0.00	0.00 ± 0.00[a]
1680	1630	0.15 + 4	Slag	50.63 ± 0.18	49.06 ± 0.41	99.7 ± 0.42	54.49 ± 0.24	45.51 ± 0.24
			Cristobalite	100.82 ± 0.15	0.00 ± 0.00[a]	100.8 ± 0.15	100.00 ± 0.00	0.00 ± 0.00[a]
1655	1650	0.15 + 4	Slag	51.92 ± 0.11	46.94 ± 0.31	98.9 ± 0.32	56.11 ± 0.16	43.89 ± 0.16
			Cristobalite	101.69 ± 0.22	0.00 ± 0.00[a]	101.7 ± 0.22	100 ± 0.00	0.00 ± 0.00[a]
no	1651	1	Slag	52.50 ± 0.26	47.50 ± 0.26	100.2 ± 0.37	56.21 ± 0.26	43.79 ± 0.26
			Cristobalite	101.16 ± 0.05	0.00 ± 0.00[a]	101.2 ± 0.05	100 ± 0.00	0.00 ± 0.00[a]
1665	1660	0.15 + 2	Slag	53.14 ± 0.42	45.97 ± 0.36	99.1 ± 0.54	57.35 ± 0.28	42.65 ± 0.28
			Cristobalite	101.12 ± 0.54	0.00 ± 0.00[a]	101.1 ± 0.54	100.00 ± 0.00	0.00 ± 0.00[a]
1675	1670	0.15 + 4	Slag	55.06 ± 0.39	43.41 ± 0.40	98.5 ± 0.55	59.50 ± 0.26	40.50 ± 0.27
			Cristobalite	100.42 ± 0.03	0.00 ± 0.00[a]	100.4 ± 0.03	100 ± 0.00	0.00 ± 0.00[a]
1695	1690	0.083 + 3	Slag	96.94 ± 1.05	3.72 ± 0.12	100.7 ± 1.04	97.17 ± 0.11[b]	2.83 ± 0.11[b]
			Cristobalite	97.36 ± 0.51	0.00 ± 0.00[a]	97.36 ± 0.51	100 ± 0.00	0.00 ± 0.00[a]
1705	1700	0.083 + 4	Slag	98.90 ± 0.25	2.67 ± 0.14	101.6 ± 0.28	98.08 ± 0.11[b]	1.92 ± 0.11[b]
			Cristobalite	99.37 ± 1.00	0.00 ± 0.00[a]	99.4 ± 1.00	100 ± 0.00	0.00 ± 0.00[a]
2 Immiscible Liquid Slags								
1680	1675	0.15 + 4	Low SiO_2 slag[c]	55.17 ± 0.84	42.89 ± 0.77	98.1 ± 0.44	59.85 ± 0.72	40.15 ± 0.72
			High SiO_2 slag[c]	95.10 ± 0.37	4.18 ± 0.10	99.3 ± 0.37	96.43 ± 0.07	3.57 ± 0.08
1685	1680	0.15 + 4	Low SiO_2 slag[c]	96.27 ± 0.48	4.41 ± 0.18	100.7 ± 0.48	59.91 ± 0.53	40.09 ± 0.53
			High SiO_2 slag[c]	56.42 ± 0.63	43.73 ± 0.62	100.2 ± 0.43	96.28 ± 0.13	3.72 ± 0.13

[a] The concentration of the element was measured using its L_α X-ray.
[b] Fe droplets were finely dispersed through the slag; additional correction for secondary fluorescence was made by subtracting the normalised molar fraction of FeO measured in cristobalite (SiO_2) using Kα X-rays from the normalised molar fraction of FeO measured in slag.
[c] The slag was metastable as it did not contain cristobalite, and a SiO_2 concentration gradient was observed near the substrate/slag interface.

3.1. "$CuO_{0.5}$"-"FeO"-SiO_2 system in equilibrium with metallic copper

The liquidus temperatures and the slag compositions of the tridymite and cristobalite (SiO_2) primary phases were measured between 1400 and 1663 °C, and the Cu/(Cu+Fe) molar ratios in the slags 0.1–0.9.

The experimental data indicated that at ≥ 1600 °C, the slag with the Cu/(Cu+Fe) = 0.5 molar ratio in equilibrium with copper metal and cristobalite had higher SiO_2 solubility than the slags with the Cu/(Cu+Fe) molar ratio 0 or 1 (the "FeO"-SiO_2 and "$CuO_{0.5}$"-SiO_2 systems in equilibrium with cristobalite, and metallic iron and copper respectively). This difference increased at higher temperatures. The slag with the Cu/(Cu+Fe) = 0.5 molar ratio contained ~70 mol.% SiO_2 at 1640 °C, compared to the slags with the Cu/(Cu+Fe) molar ratios 0 and 1 which contained ~55 mol.% SiO_2 and ~35 mol.% SiO_2 respectively.

The two liquid slags miscibility gap over cristobalite was characterised by tie lines connecting a low-SiO_2 and a high-SiO_2 liquid slag with the approximately equal Cu/(Cu+Fe) ratios. This miscibility gap was found to be present at all Cu/(Cu+Fe) molar ratios in the slag. However, the difference in the SiO_2 concentration between the low- and high-SiO_2 slag was found to decrease as the Cu/(Cu+Fe) molar ratio in the slag approached 0.5 from either 0 or 1. The minimum difference in the SiO_2 concentration between the low- and high-SiO_2 slag was assessed to be approximately 4 mol.%, at the Cu/(Cu+Fe) molar ratio in slag = 0.5. This was attributed to the charge compensation effect [37]. Fe^{3+} cations replaced the Si^{4+} cations in the tetrahedral position of the slag network surrounded by four O^{2-} anions, with the excess negative charge compensated by the nearby Cu^{+} cations in the interstitial sites of the tetrahedral network. The solutes formed strong covalent bonds with the silicate slag network as a result. This may explain the increased stability of the liquid slag with the Cu/(Cu+Fe) ≈ 0.5 molar ratio, and the observed trend in the solubility of SiO_2 in the "$CuO_{0.5}$"-"FeO"-SiO_2 slag.

3.2. "FeO"-SiO_2 system in equilibrium with metallic iron

The liquidus temperatures and the slag compositions of the tridymite and cristobalite (SiO_2) primary phase fields in the "FeO"-SiO_2" system in equilibrium with metallic iron were measured between 1450–1680 °C, and 40–95 mol.% SiO_2 in the slag.

The liquid slag composition measured in the present study agreed with the results by Bowen and Schairer [17] at 1450 °C. However, the tridymite and cristobalite (SiO_2) liquidus measured in the present study was less steep than that reported by the earlier studies [17,18]. The present study measured the slag in equilibrium with cristobalite as containing 5 mol.% SiO_2 higher than the slag measured by Turkdogan and Bills [18] at 1600 °C.

The monotectic temperature was measured by performing a dedicated series of experiments, during which the equilibration temperature was decreased in 5 °C increments until the high-SiO_2 slag was replaced with cristobalite (SiO_2). The low- and high-SiO_2 slags were measured in the present study as containing 59.7 mol.% and 96.4 mol.% SiO_2 respectively at the monotectic temperature of 1673 ± 3 °C. This differed from the results by Greig [16], who measured the SiO_2 concentrations to be 60.6 mol.% and 97.4 mol.% respectively at the monotectic temperature of 1689 °C. These discrepancies were likely due to the limitations of the equilibration experiments conducted by Greig [16]. Separate experiments were used in their work to determine the compositions of the low- and

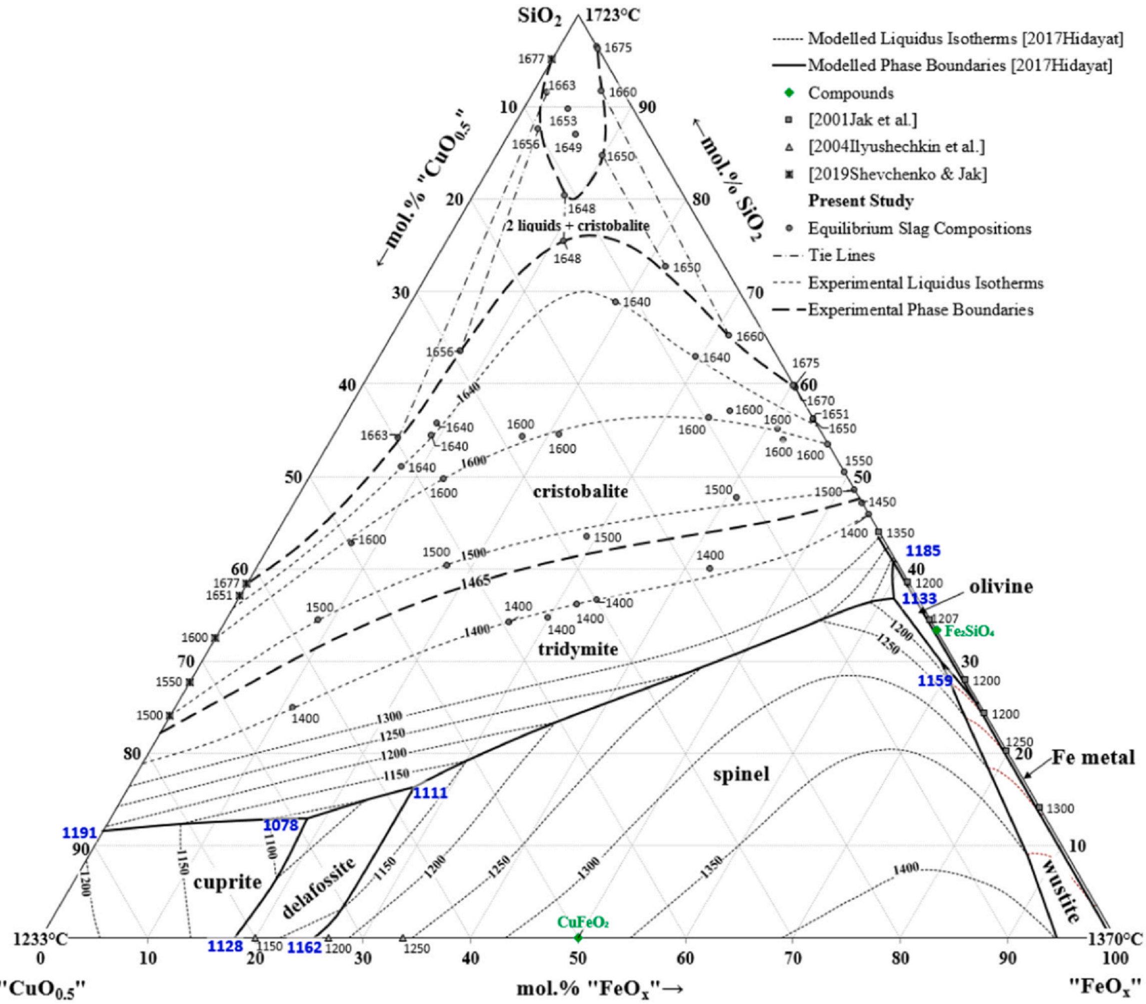

Fig. 4. Liquidus surface of the "$CuO_{0.5}$"-"FeO_x"-SiO_2 pseudo-ternary system in equilibrium with metal. Phase boundaries and isotherms at > 1400 °C were constructed using the experimental results from the present study, while the remainder of the diagram was calculated using the thermodynamic model by Hidayat et al. [19,20]. Experimental data from the present study, and by Ilyushechkin et al. [8], Shevchenko and Jak [15], and Jak et al. [40] were included for reference.

high-SiO_2 immiscible slags at the monotectic temperature, with low proportions of the high- and low-SiO_2 slags present in the final samples respectively. Additionally, the equilibration times used by Greig [16] were shorter (5–10 min) and increasing equilibration times were associated with the increased Fe oxidation state due to impurities in their gas atmosphere.

The experimentally measured slag compositions and temperature at the monotectic point differed significantly from the latest thermodynamic assessment of the Fe-Si-O system by Hidayat et al. [20], who modelled the low- and high-SiO_2 slags as containing 57.9 mol.% and 97.7 mol.% SiO_2 respectively at the monotectic temperature of 1686 °C.

3.3. Solubility of "$CuO_{0.5}$" and "FeO_x" in tridymite and cristobalite (SiO_2)

When the concentrations of "$CuO_{0.5}$" and "FeO_x" in tridymite and cristobalite (SiO_2) were measured using their respective K_α X-rays during the present study, the concentration of "$CuO_{0.5}$" was measured to be between 0.23 and 0.63 mol.% in tridymite and to be between 0.09 and 1.3 mol.% in cristobalite, while the concentration of "FeO_x" was measured to be between 0.43 and 0.61 mol% in tridymite, and to be between 0.05 and 0.71 mol% in cristobalite. These were consistent with the apparent solubilities of "$CuO_{0.5}$" and "FeO_x" in tridymite and cristobalite (SiO_2) reported in previous studies of the "$CuO_{0.5}$"-"FeO"-SiO_2 slag systems, when the composition was measured using EPMA with Kα X-rays [5,10,11,14,15,28,30,38,39]. A summary of the measured "$CuO_{0.5}$" and "FeO_x" concentrations reported in those studies is given in Table 2.

By comparison, the concentration of "$CuO_{0.5}$" was measured to be between 0 and 0.04 mol.% in tridymite and between 0 and 0.03 mol.% in cristobalite (SiO_2), while the concentration of "FeO_x" was consistently measured to be 0.00 mol% when measured using their respective L_α X-rays. The measured concentration of metallic impurities in tridymite did not exhibit any trend with the Cu/(Cu+Fe) ratio, indicating that nonzero apparent solubilities were likely noise. Thus, it was concluded that the apparent concentrations of "$CuO_{0.5}$" and "FeO_x" in tridymite and cristobalite (SiO_2) were overestimated by measurements with the K_α X-rays, and that their true solubilities were lesser than or equal to the lower detection limits of "Cu" and "Fe" reported in Table 1, multiplied by the quotient between the molar mass of their respective oxide (normalised to one mole of metallic cation) and the molar mass of the metal.

A key limitation of the EPMA measurements made in the present study was the relatively short background measurement times used. This was an issue especially when measuring Fe concentration in SiO_2; due the low peak intensity, the lower detection limit of Fe in SiO_2 when measured with the Lα X-rays does not preclude the possibility of Fe solubility in tridymite or cristobalite (SiO_2) to the order of below 0.2 wt%. To reduce the detection limit of Fe and Cu in tridymite and cristobalite (SiO_2) in future studies, it is recommended

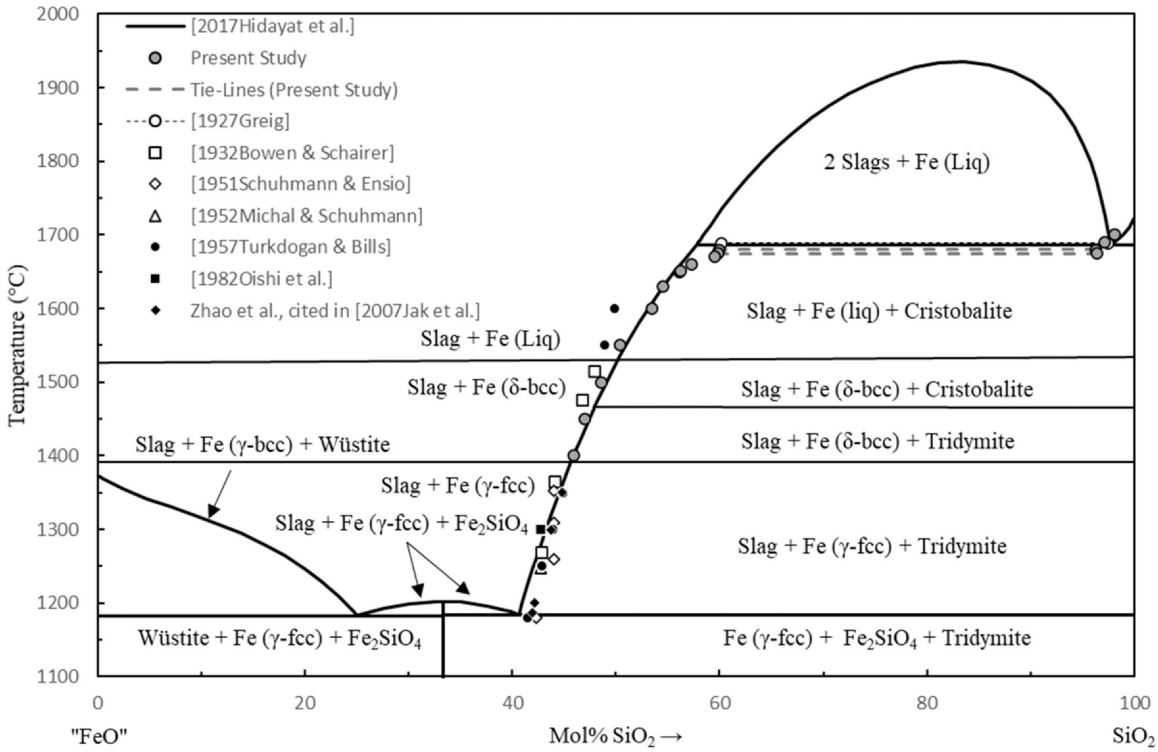

Fig. 5. Liquidus surface of the “FeO”-SiO2 pseudo-binary system in equilibrium with metal. The phase diagram was calculated using the thermodynamic model by Hidayat et al. [20]. Experimental data from the present study, and from previous studies [2,16–18,41–43] were included for reference.

that either the background measurement time or the probe current are increased for measurements with the Lα X-rays.

4. Conclusions

The range of conditions studied for the “$CuO_{0.5}$”-“FeO”-SiO_2 slag system in equilibrium with metallic copper was extended in the present study to include temperatures from 1400 °C to the monotectic temperature of two immiscible liquids over cristobalite, in the direction of increasing SiO_2 concentrations in slag (up to 95 mol.%). Although the two-liquids miscibility gap was found to be present at all Fe/(Cu+Fe) molar ratios in the slag, charge compensation between Cu^+ and Fe^{3+} ions in the slag decreases the difference in SiO_2 concentration between the high- and low- SiO_2 slags to a minimum of 4 mol.%, at the Fe/(Cu+Fe) molar ratio = 0.5 in slag. Additionally, new phase equilibria data were obtained for the “FeO”-SiO_2 subsystem in equilibrium with metallic iron at high temperatures (from 1400 °C to two immiscible liquids over cristobalite). The new results suggested that the monotectic temperature of the subsystem is 1673 ± 3 °C, and that the low- and high- SiO_2 slags contain 59.7 mol.% and 96.4 mol.% SiO_2 respectively. Lastly, measurements of the “$CuO_{0.5}$” and “FeO” concentration in tridymite and cristobalite (SiO_2) using their Lα X-rays indicated that their true solubilities were below the lower detection limits of the EPMA, and that the much higher apparent solubilities reported in previous studies were likely due to the secondary fluorescent Kα X-rays generated in the slag adjacent to the phases. The present study was a major extension of previous investigations into the Cu-Fe-Si-O and Fe-Si-O systems, and formed part of a broader study of the multicomponent Pb-Zn-Cu-Fe-Si-Ca-Al-Mg-O system for nonferrous metal smelting and recycling industries.

CRediT authorship contribution statement

Xi Rui Wen: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **Maksym Shevchenko:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - review & editing, Supervision, Project administration, **Evgueni Jak:** Conceptualization, Validation, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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